PV Mounting Structure

Request for Quotation & Structural Design Brief
185.4 kWp rooftop PV – Marriott Hotel, mezzanine roof

Doc. PV-RFQ-STR-001 Rev A
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Ref. drawing D18102-DD-A-104

We invite your quotation for the design, supply and delivery of the PV mounting structure described below. The design must be carried out by a qualified structural engineer, and the calculations submitted for approval by the building's structural engineer.

1. Project summary

Location	Uganda (equatorial, ≈0–1° N). Site address to be given at award.
Roof	Flat reinforced-concrete slab with perimeter parapet, single level; waterproofed. Usable area ≈29.4 × 39 m.
Roof height	About 7 m above ground (mezzanine podium). **Two hotel towers stand next to the roof**; consider wind acceleration and downwash.
PV system	299 modules × 620 Wp = 185.4 kWp
Module	Bifacial TOPCon, 2382 × 1134 × 30 mm, ≈33 kg (dual glass); datasheet to follow
Design life	25 years minimum

2. Array configuration

Item	Value
Tilt / orientation	10° fixed; modules portrait, rows running along the 29.4 m roof width
Table type	2P = 2 modules high in portrait × 23 wide; 1P = 1 module high × 23 wide
Quantities	**6 tables of 2P × 23 (46 modules each) + 1 table of 1P × 23 (23 modules) = 299 modules**
Table length	26,522 mm (23 modules + 20 mm gaps)
Table depth on plan	4,711 mm (2P); 2,346 mm (1P)
Front edge height (min.)	300 mm above slab (cleaning access, water flow, bifacial gain)
Rear edge height	≈1,131 mm (2P); ≈714 mm (1P)
Gaps	20 mm between modules; 800 mm aisle between tables; ≥1,000 mm from parapet

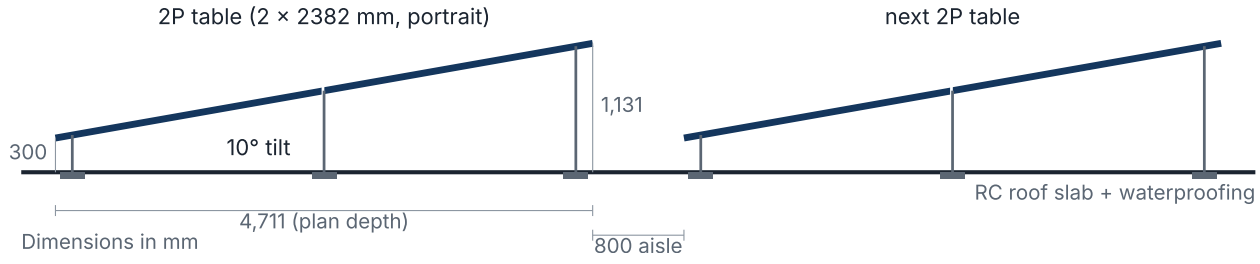


Figure 1 – Typical section through two 2P tables

3. Layout

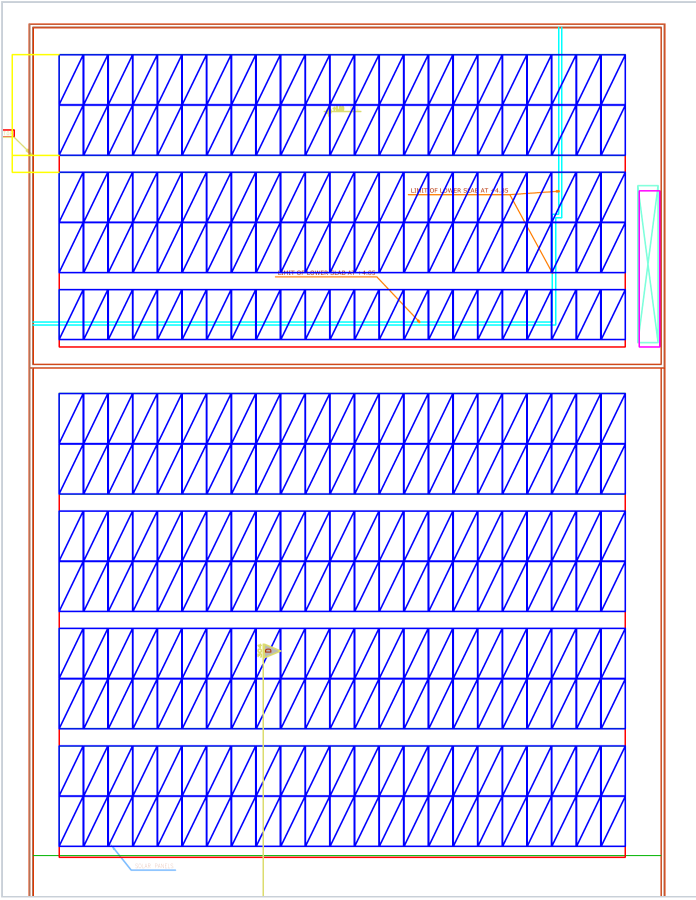


Figure 2 – PV layout on the mezzanine roof (CAD file DWG/DXF supplied with this RFQ)

4. Structural design basis

Item	Requirement / preliminary value
Design codes	EN 1990, EN 1991-1-4 (wind), EN 1999 (aluminium), EN 1993 (steel), or ASCE 7 equivalent. Seismic per Uganda code US 319.
Basic wind speed	v_b = 25 m/s (10-min mean, 10 m, preliminary). **Supplier to confirm for the site.**
Peak velocity pressure	q_p ≈ 0.67 kN/m² (terrain cat. III, z = 10 m, c_e ≈ 1.70); check tower effects
Wind pressure coefficients	From supplier's wind-tunnel data for 10° arrays, with edge and corner zones next to the parapet
Module dead load	≈12.2 kg/m² of module area (33 kg per module)
Structure dead load (allowance)	≈4.5 kg/m² of module area; supplier to state actual
Snow load	None
Maintenance load	Structure not walkable; aisles carry roof live load (by others)
Temperature range	10 °C to 70 °C (structure); expansion breaks in rails at ≤ 12 m
Deflection limit	Rails L/200; must meet the module manufacturer's limits

Corrosion	Category C3. Aluminium 6005-T6 / 6063-T6 (anodised or mill finish); fasteners stainless A2-70 / A4-70; no galvanic contact with steel
Module clamping	Long-side clamping within the zones of the module installation manual (dual-glass bifacial); clamp loads to rated test loads (5400 Pa down / 2400 Pa up)

5. Roof interface

- **Option A – Anchored (preferred):** feet fixed to concrete plinths or chemical anchors into the slab, with full waterproofing detail. Suits the wind exposure near the towers.
- **Option B – Ballasted:** no penetrations; concrete ballast blocks with wind deflectors. Must state ballast per table and the resulting load in kg/m².
- EPDM or rubber protection pads under every foot; do not block roof drainage paths or outlets.
- **Allowable roof load to be confirmed by the building's structural engineer.** The supplier must give the uniform load (kg/m²) and maximum point load (kN) at each foot.

Indicative weights: modules 9.9 t + structure ≈3.6 t (excluding ballast) over ≈812 m² of table footprint, i.e. ≈17 kg/m² before ballast.

6. Scope of supply

- Rails, rafters, legs, bracing for 7 tables
- End and mid clamps, bonding/earthing clips
- All fasteners (stainless steel)
- Feet, anchors or ballast trays, protection pads
- Optional: roof walkway between tables
- Optional: cable trays fixed to the structure
- Optional: installation supervision

7. Information required with the quotation

1. Price per Wp and total price, in USD; state delivered to site or ex-works, and lead time.
2. Structural calculation report: wind zones, loads per foot (kN), ballast or anchor details, slab load (kg/m²).
3. General arrangement and detail drawings, with a bill of materials.
4. Material grades, corrosion protection and certificates.
5. Warranty period (structure 10 years minimum; materials 20 years minimum) and installation manual.

Contact: [Name / Company / Phone / Email]